CHAO LEI

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Personal Homepage: https://you68681.github.io/

Research Profile

My research focuses on reasoning, planning, and generalisation in large language models (LLMs), multimodal large language models (MLLMs), and agentic systems. I develop reinforcement learning, knowledge-augmented, and planning-based methods to improve reasoning, decision-making, and generalisation in LLMs and MLLMs. My previous work spans state-aware multimodal reasoning, program synthesis, Theory of Mind, and automated planning. My current research focuses on multi-agent reinforcement learning (Multi-Agent RL) and multi-agent recursive self-improvement (RSI), with an emphasis on agent cooperation, long-horizon planning, and adaptive reasoning.

Academic Background

PhD in Computer Science, The University of Melbourne

2022 – Present | Thesis submitted in July 2026; degree expected in December 2026.

Research Topic: Improving Abstraction and Generalization in Reasoning Systems

Supervisors: A/Prof Nir Lipovetzky, A/Prof Kris Ehinger

Postgraduate

2019 – 2020 Master of Computer Science (The University of Melbourne)

2017 – 2018 Master of Information Technology (The University of Melbourne)

Undergraduate

2012 – 2016 Bachelor of Aircraft Design and Engineering (Northwestern Polytechnical University, China)

Publications

✧ **SVoT: State-aware Visualization-of-Thought for Spatial Reasoning via Reinforcement Learning**

C. Lei, Y. Jiang, M. Hiller, Z. Zhou, X. Tian, K. A. Ehinger, N. Lipovetzky. The Fortieth Annual Conference on Neural Information Processing Systems, 2026 (NeurIPS).

✧ **From Reasoning to Generalization: Knowledge-Augmented LLMs for ARC Benchmark**

C. Lei, K. A. Ehinger, N. Lipovetzky, Y. Chang. The Conference on Empirical Methods in Natural Language Processing, 2026 (EMNLP Main Track).

✧ **Mind the Perspective: Let's Reason Recursively for Theory of Mind**

C. Lei, G. Hu, M. Yang, Y. Jiang, N. Lipovetzky. Findings of the Conference on Empirical Methods in Natural Language Processing, 2026 (EMNLP Findings).

✧ **MIDI-LLaMA: An Instruction-Following Multimodal LLM for Symbolic Music Understanding**

M. Yang, J. McCormack, M. T. Llano, W. Su, **C. Lei**. In Proceedings of the International Conference on Acoustics, Speech, and Signal Processing, 2026 (ICASSP)

✧ **DUAL: Learning Diverse Kernels for Aggregated Two-sample and Independence Testing**

Z. Zhou, X. Tian, L. Peng, **C. Lei,** A. Schrab, D. J. Sutherland, F. Liu. In Proceedings of the 39th Annual Conference on Neural Information Processing Systems, 2025 (NeurIPS)

✧ **Planning-Driven Programming: A Large Language Model Programming Workflow**

C. Lei, Y. Chang, N. Lipovetzky, K. A. Ehinger. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, 2025 (ACL Main Track)

✧ **Beyond Perception: Evaluating Abstract Visual Reasoning through Multi-Stage Task**

Y. Jiang, Y. Ding, **C. Lei,** J. Ao, J. Lau, K. A. Ehinger. In Findings of the ACL, 2025 (ACL Findings)

✧ **State-Based Disassembly Planning**

C. Lei, N. Lipovetzky, K. A. Ehinger. In Proceedings of the AAAI Conference on Artificial Intelligence, 2025 (AAAI)

✧ **Generalized Planning for the Abstraction and Reasoning Corpus**

C. Lei, N. Lipovetzky, K. A. Ehinger. In Proceedings of the AAAI Conference on Artificial Intelligence, 2024 (AAAI)

✧ **Novelty and Lifted Helpful Actions in Generalized Planning**

C. Lei, N. Lipovetzky, K. A. Ehinger. In Proceedings of the International Symposium on Combinatorial Search, 2023 (SoCS)

✧ **Width-Based Backward Search**

C. Lei and N. Lipovetzky. In Proceedings of the International Conference on Automated Planning and Scheduling, 2021 (ICAPS)

Preprints

✧ **CARE: Grounded Recursive Self-Improvement for Production Customer-Service Agents**

Y. Jiang, **C. Lei**, R. Gao, X. Zhao, Y. Ge, H. Liu, R. Nie, et al. arXiv preprint, 2026

✧ **PROPA: Toward Process-level Optimization in Visual Reasoning via Reinforcement Learning**

Y. Jiang, **C. Lei,** Y. Ding, K. A. Ehinger, J. Lau. arXiv preprint, 2026

✧ **PathISE: Learning Informative Path Supervision for Knowledge Graph Question Answering**

S. Gao, **C. Lei**, J. H. Lau, J. Qi. arXiv preprint, 2026

Work Experience

✧ Research Assistant, The University of Melbourne (Feb 2026-Present)
- ➢ Working with Melbourne School of Design on automated EPiC database generation with LLMs.

✧ Research Assistant, The University of Melbourne (Sep 2025-Dec 2025)
- ➢ Worked with Google on automated fuzz driver generation with LLMs.

✧ Tutor, AI Planning and Autonomy, The University of Melbourne (July 2024 – Present)
- ➢ Tutored planning algorithms, reinforcement learning algorithms, and game theory.

✧ Tutor, Computer Vision, The University of Melbourne (Feb 2025 –July 2025)
- ➢ Tutored computer vision algorithms for object recognition, detection, and 3D reconstruction.

✧ Web Development for CORE Ranking, RMIT University (Part-Time, Feb 2019 – Feb 2020)
- ➢ Full-stack web development for the CORE Ranking platform using JavaScript, Python, and MySQL.

Master's Research Project Supervision

✧ Jikun Yang. Thesis: **Applying Generalized Planning to Non-Markovian Problems (**2024-2025**)**

✧ Liao Cao. Thesis: **Generalized Planning in Epistemic Planning with Justified Perspective (**2024-2025**)**

✧ Dajun Guo. Thesis: **Planning-Augmented Text-to-Image Generation in MLLMs** (2025-2026)

✧ Xinran Wang. Thesis: **Memory Management for Large Language Model Agents** (2026-Present)

Invited Talks

✧ University of Southern California. Topic: **Novelty in Generalized Planning** (2023)

✧ University of Toronto. Topic: **Generalized Planning for the Abstraction and Reasoning Corpus** (2023)

✧ Oracle (Melbourne). Topic: **Programming with LLMs: A Planning-Driven Approach** (2025)

Academic Service

✧ **Program Committee Member:** ICAPS, AAAI, CVPR, ICCV, ECCV, ICML, NeurIPS

✧ **Conference Chair Roles:** ICAPS 2025 Publicity Chair

Awards and Scholarships

✧ SoCS Travel Scholarship 2023

✧ Melbourne Research Scholarship, The University of Melbourne (2022-2025)

✧ Second Prize, Undergraduate Mathematical Contest in Modeling, Shaanxi Division, 2017 and 2018.